Problem determination and troubleshooting

How to perform problem determination actions on the water-cooling module

Troubleshooting overview

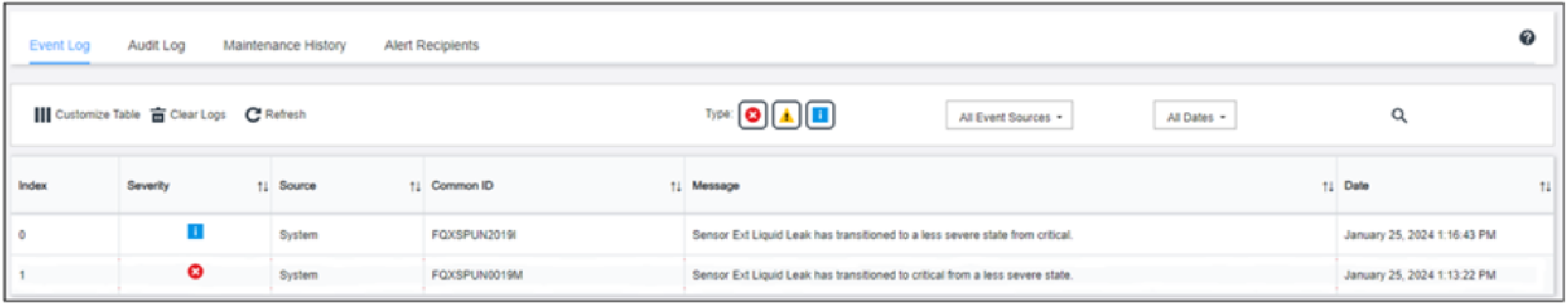
To determine the cause of problems with Neptune Liquid Cooling solutions, work through the following actions:

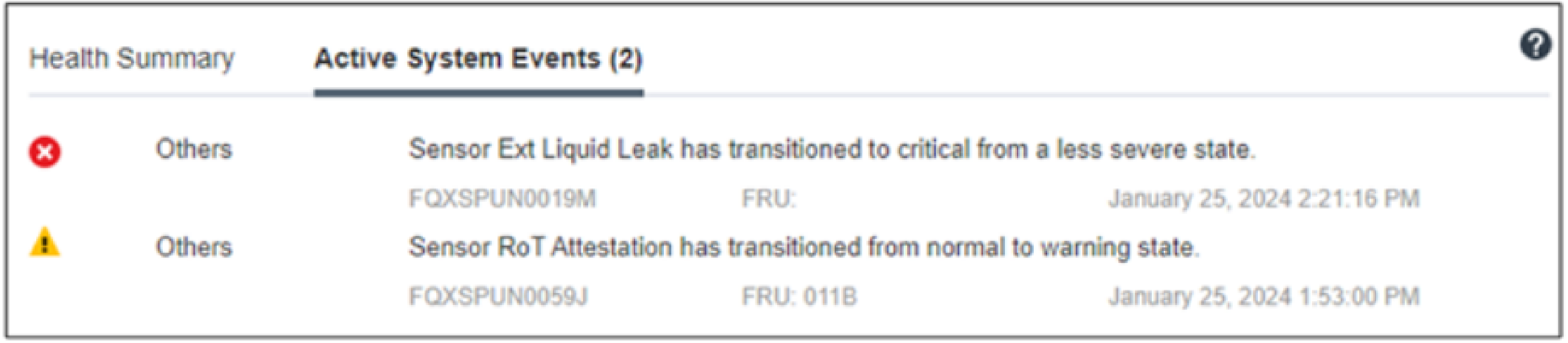
- Check the system status and the system event log on the XCC dashboard. Issues with water cooling modules might cause high temperatures in the processor, GPU, and other major components.
- Check the operating panel on the front of the system.
- Inspect the LEDs on the leakage detection sensor and the LED on the water-cooling module.
- For the DW612 or DW612S enclosures, check the drip sensor LED.
- Visually check if there is a coolant leak from the water-cooling module and manifold.
- For the ThinkSystem SD650, check the enclosure FDC messages.
- For the ThinkSystem SD650 V2, SD650 V3, and SD665 V3, check the enclosure SMM2 messages.
- Use IPMI commands to identify a coolant leak.

Note: For detailed error messages of water leakage problems, go to Lenovo Docs.

Using XCC to identify a coolant leak

If the amber LED on the front operator panel is lit, check the XCC event log.
If an XCC event has an ID of **FQXSPUN0019M** and a message stating: **Sensor Liquid Leak has transitioned to critical from a less severe state**, it can be identified as a coolant leakage.



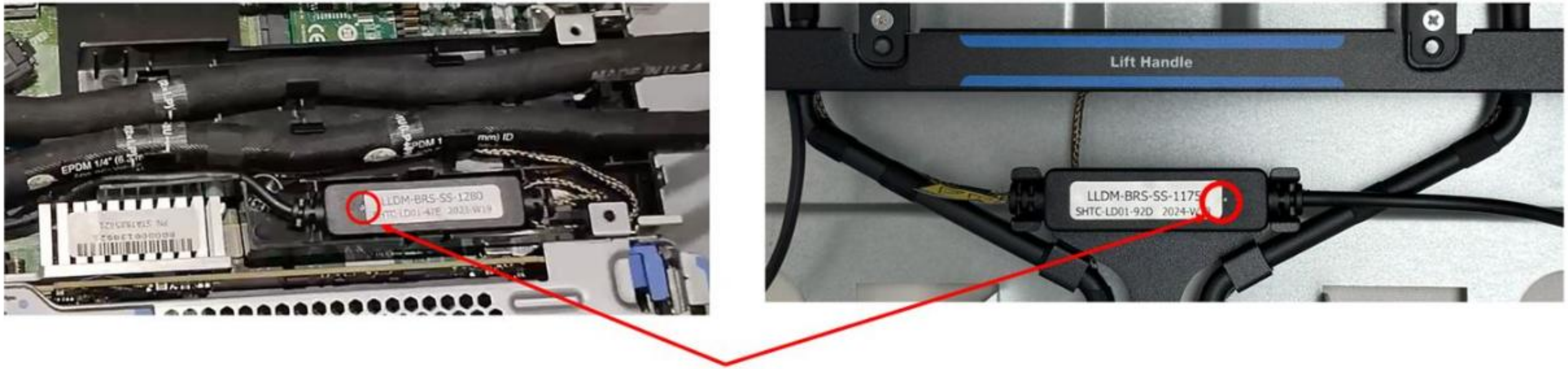


Leakage detection sensor module

The leakage detection sensor module is used to detect coolant leakages in the system. When the module is installed in the hose holder, the leakage detection sensor LED should face up. A blinking green light indicates an abnormal status, while a solid green light indicates that no coolant leakage has been detected.

DWCM

L2AM

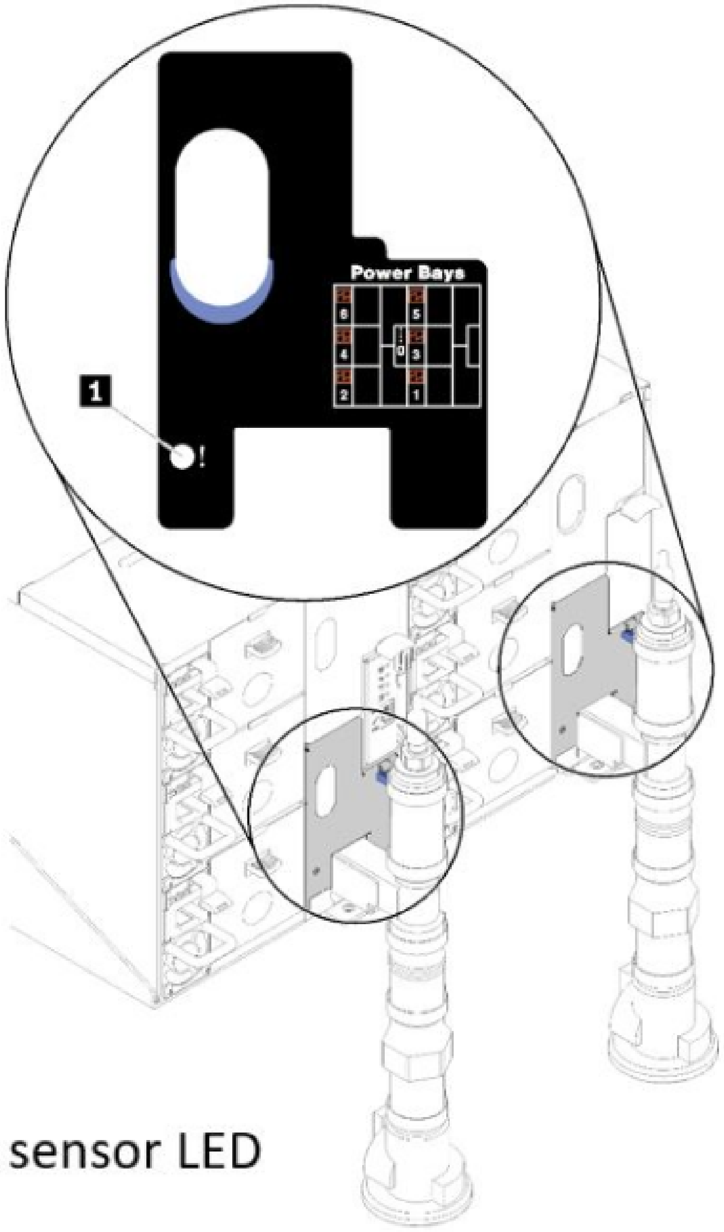


Leakage detection sensor LED

DW612 and DW612S drip sensor module

The DW612 drip sensor is used in both the DW612 and DW612S to monitor enclosure leaks. The sensor is designed to detect drips or leaks from the liquid-cooling system that could cause damage to the server or its components. Each enclosure has two drip sensors at the rear of the system.

> **Note:** If a sensor detects moisture in its drip sensor catch basin, it activates a yellow LED that is visible through a hole located in the lower left leg of the lower EMC shield.



1 Drip sensor LED

Using IPMI commands to identify a coolant leak

If the amber LED is lit on the front operator panel, IPMI commands can be used to check the system running status. IPMI commands with a `sel elsit` parameter can be used to check for a coolant leakage status.

```
zuody2@zuody2-07:~$ ipmitool -H 10.245.50.35 -U USERID -P Aa12345678 -I lanplus -C 17 sel elist
   1 | 01/25/2024 | 13:40:30 | Event Logging Disabled SEL Fullness | Log area reset/cleared | Asserted
   2 | 01/25/2024 | 13:41:58 | Cooling Device Ext Liquid Leak | Transition to Critical from less severe | Asserted
   3 | 01/25/2024 | 13:42:01 | Cooling Device Ext Liquid Leak | Transition to Critical from less severe | Deasserted
```

If there is a coolant leakage, the log indicated above will be displayed. The status of all sensors can be checked with the `sdr elist` parameter. (Click HERE for a picture.)

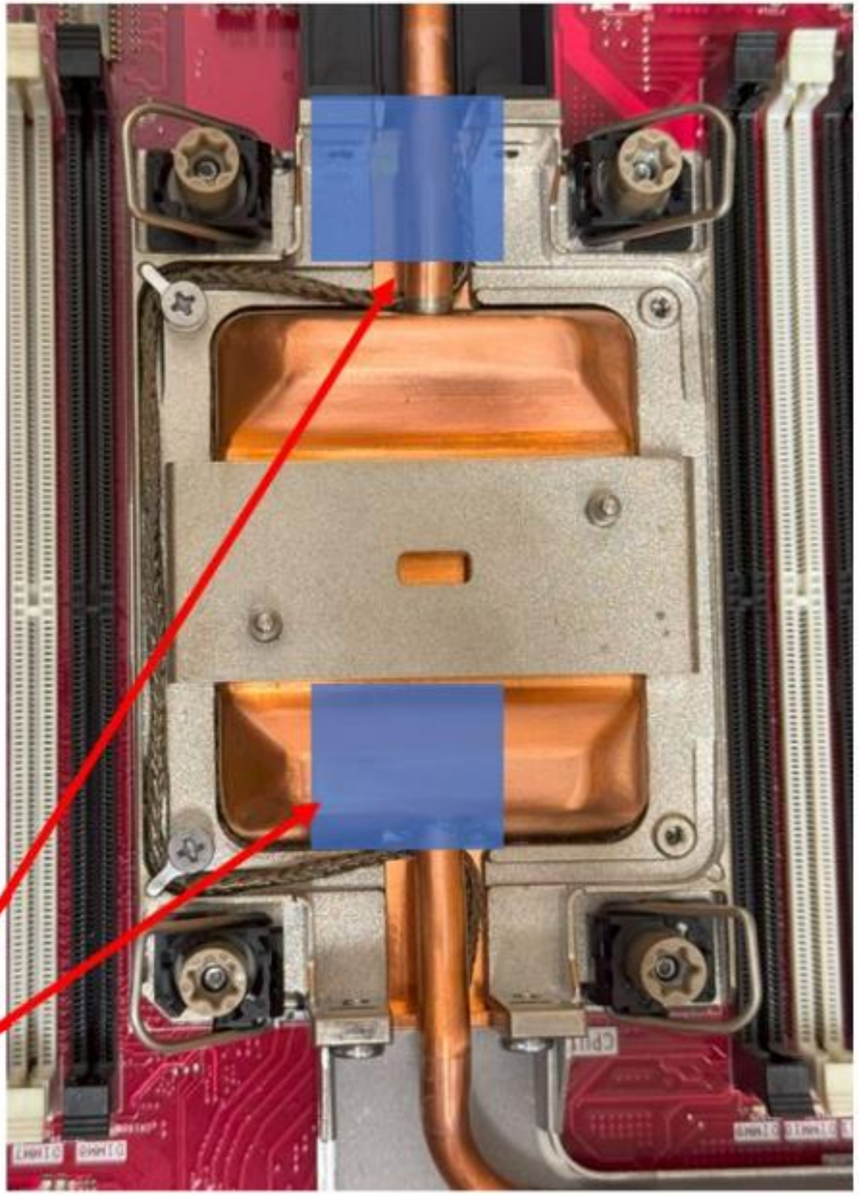
```
C:\ipmitool>ipmitool -I lanplus -H 192.168.70.125 -U USERID -P xccPASSWORD sdr elist
CPUs             | 0Ah | ok  | 37.3 |
All CPUs         | 04h | ok  | 37.3 |
One of CPUs      | 18h | ok  | 37.3 |
CPU 1            | F2h | ok  |  3.1 | Presence detected
CPU 2            | F3h | ok  |  3.2 | Presence detected
CPU 1 UPILinkErr | F4h | ok  |  3.1 |
CPU 2 UPILinkErr | F5h | ok  |  3.2 |
Sys Boot Status  | 24h | ok  | 34.1 |
Sys Utilization  | DCh | ok  | 19.1 | 0 unspecified
CPU Utilization  | D9h | ok  | 19.1 | 0 unspecified
Mem Utilization  | DBh | ok  | 19.1 | 0 unspecified
IO Utilization   | DAh | ok  | 19.1 | 0 unspecified
ME Status        | F8h | ok  | 46.1 |
Sys Power        | A1h | ok  | 19.1 | 20 Watts
CPU Power        | A3h | ns  | 19.2 | No Reading
Mem Power        | A4h | ns  | 19.3 | No Reading
PSU Mismatch     | FBh | ok  | 19.1 | Transition to OK
PS Heavy Load    | FCh | ok  | 19.1 | State Deasserted
Power Resource   | FDh | ok  | 19.1 |
Power Supply 1   | DDh | ok  | 10.1 | Presence detected
PSU1 Failure     | DEh | ok  | 10.1 | Transition to OK
PSU1 PF Failure  | DFh | ok  | 10.1 | Transition to OK
PS[illegible] IN Failure  | E0h | ok  | 10.1 | Transition to OK
PSU1 AC In Pwr   | A5h | ok  | 19.5 | 224 Watts
```

Resolving DWCM coolant leaks

If the LED on the leakage detection sensor module is blinking green or if XCC reports a leakage event, work through the following steps to resolve the problem:

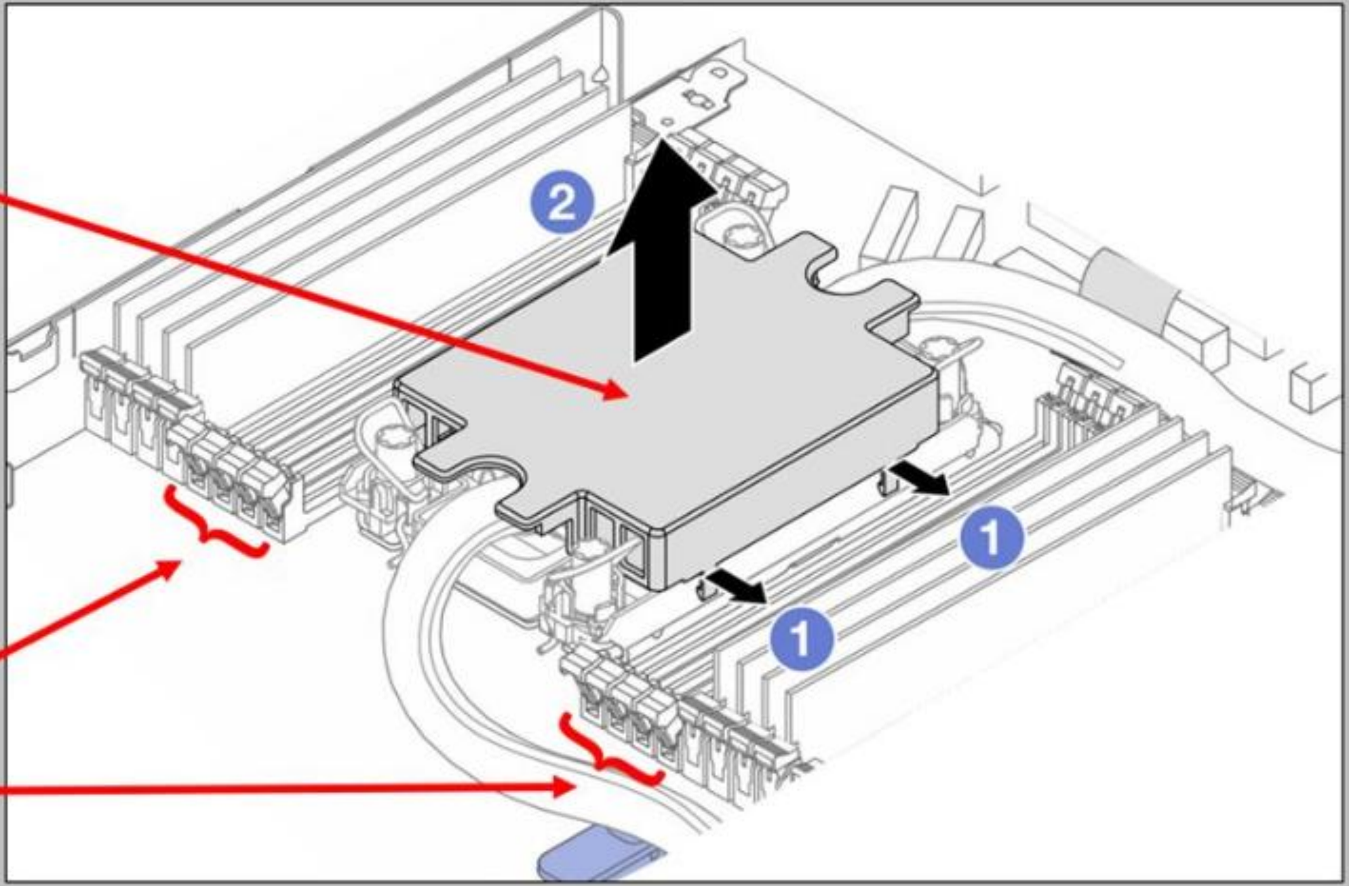
- Save and back up data and server operations.
- Power off the server and remove the quick connect plugs from the manifolds.
- Slide out the server or remove it from the rack, and then remove the top cover.
- Check for coolant leaks around the outlet and inlet hoses, system board assembly, and under the cold plate covers.
- If coolant is found around the hoses and system board assembly, clean it up.
- If coolant is found under the cold plate covers, remove at least four DIMMs on each side to get access to the clips on the cold plate covers. (Click HERE to see a picture.)
- Replace the DWCM.
- As coolant might have dripped down on to the lower servers, check the top cover of the next server down. If you find coolant, work through these steps again.

Note: If there is a leak, the coolant tends to collect in leak-prone areas.



Resolving DWCM coolant leaks

If there is a coolant link, remove the cold plate cover.



Note: Remove at least four DIMMs from each side to get access to the clips on the cold plate covers.

Resolving L2AM coolant leaks

If the LED on the leakage detection sensor module is blinking green or if XCC reports a leakage event, work through the following steps:

- Check for green coolant leakage around the radiator, coolant pipes, and processor cold plate.
- If green coolant is found, power off the server and remove the liquid to air module.
- Clean up the coolant from any components in the chassis, and then inspect the server for any signs of moisture in sockets or gaps.
- Replace the liquid to air module.